Project Report
On
Computer Networking: Concepts
(CSE3751)

University Network Design & Implementation

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B. Tech. BRANCH 5th Semester (Section --- 23412-A2)

INSTITUTE OF TECHNICAL EDUCATION AND RESEARCH

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Declaration

We, the undersigned students of B. Tech. of **(CSE)** Department hereby declare that we own the full responsibility for the information, results etc. provided in this PROJECT titled "**University Network Design & Implementation**" submitted to **Siksha 'O' Anusandhan (Deemed to be University), Bhubaneswar** for the partial fulfillment of the subject **Computer Networking: Concepts (CSE 3751)**. We have taken care in all respects to honor the intellectual property right and have acknowledged the contribution of others for using them in academic purpose and further declare that in case of any violation of intellectual property right or copyright we, as the candidate(s), will be fully responsible for the same.

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Abstract

This project focuses on the design and implementation of a campus-wide university network using Cisco Packet Tracer. A three-tier hierarchical architecture consisting of Core, Distribution, and Access layers is implemented to ensure scalability, performance, and manageability. VLAN-based segmentation is used to isolate departments such as CSE, ECE, Mechanical, Civil, Administration, Library, and Server Farm. Inter-VLAN routing, DHCP, DNS, Web, Email, and FTP services are configured to simulate real-world enterprise networking. OSPF is used for internal routing, NAT is configured for Internet access, and ACLs are applied to secure sensitive VLANs. The project demonstrates successful connectivity, service accessibility, and security enforcement through testing and verification.

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1. Introduction

In today's digital era, universities and educational institutions heavily depend on computer networks to support a wide range of academic, administrative, and research activities. From online learning platforms, digital libraries, and examination systems to email communication, file sharing, and intranet portals, a reliable and secure network infrastructure has become a fundamental requirement. As the number of users and services increases, the network must be scalable, well-structured, and capable of handling large volumes of data traffic without performance degradation.

A poorly designed network can lead to frequent downtime, security vulnerabilities, inefficient resource utilization, and difficulty in future expansion. Therefore, modern campus networks are typically designed using hierarchical models, segmentation techniques, and standardized routing protocols. The three-tier hierarchical model—comprising Core, Distribution, and Access layers—is widely adopted in enterprise and campus environments due to its modular design, ease of troubleshooting, and scalability. This model allows network administrators to separate responsibilities such as high-speed backbone routing, policy enforcement, and end-device connectivity.

This project focuses on the design and implementation of a campus-wide university network using Cisco Packet Tracer, a widely used network simulation and visualization tool. The simulated university network includes multiple academic departments such as Computer Science and Engineering (CSE), Electronics and Communication Engineering (ECE), Mechanical Engineering, Civil Engineering, along with Administration, Library, and a centralized Server Farm. Each department is logically separated using Virtual Local Area Networks (VLANs) to enhance security, reduce broadcast traffic, and improve overall network performance.

To enable communication between these isolated VLANs, inter-VLAN routing is implemented using a core routing device. Dynamic IP address allocation is achieved through a centralized DHCP server, ensuring efficient IP management and reduced manual configuration. Essential campus services such as Domain Name System (DNS), Web (Intranet), Email (SMTP/POP3), and File Transfer Protocol (FTP) are hosted within the server farm to simulate real-world enterprise services commonly found in university environments.

Routing within the campus backbone is achieved using the Open Shortest Path First (OSPF) protocol, which provides fast convergence and efficient route selection. Internet connectivity is simulated through an edge router using Network Address Translation (NAT), allowing internal private IP addresses to access external networks securely. Additionally, Access Control Lists (ACLs) are configured to restrict unauthorized access to sensitive VLANs such as Administration and Server networks, thereby enhancing network security.

Overall, this project provides hands-on exposure to core computer networking concepts including VLANs, routing protocols, network services, security mechanisms, and hierarchical network design. The implementation closely resembles a real-world university campus network and serves as a practical learning experience for understanding enterprise-level networking using Cisco Packet Tracer.

2. Problem Statement

Universities today operate as large-scale enterprise environments where thousands of users, including students, faculty members, administrative staff, and visitors, access network resources simultaneously. These users require uninterrupted access to academic portals, internal communication systems, digital libraries, and centralized servers. As a result, the network must be carefully designed to ensure efficiency, scalability, security, and reliability.

The objective of this project is to design and implement a fully functional university campus network that interconnects multiple academic departments and administrative units while maintaining logical separation and controlled access. The network must support departmental independence through segmentation while still allowing necessary inter-department communication for shared resources and services.

The proposed university network should incorporate the following requirements:

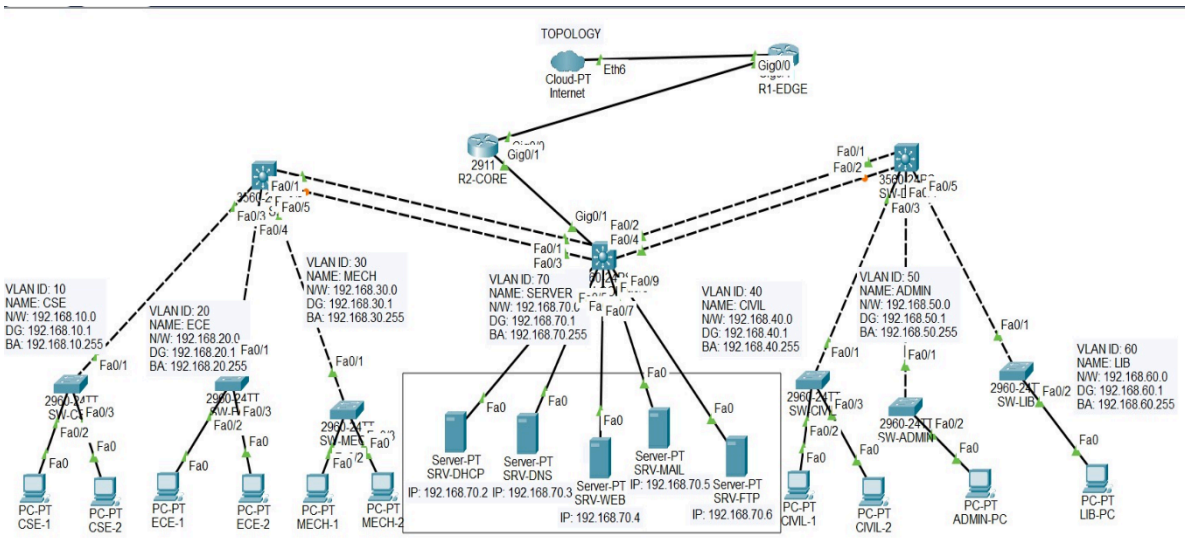
- Logical segmentation of departments such as CSE, ECE, Mechanical, Civil, Administration, Library, and Server Farm using Virtual Local Area Networks (VLANs) to reduce broadcast traffic and improve security.
- Implementation of inter-VLAN routing to enable controlled communication between different departments and shared services.
- Deployment of centralized network services including Dynamic Host Configuration Protocol (DHCP) for automatic IP address assignment, Domain Name System (DNS) for hostname resolution, Web server for intranet access, Email server for internal communication, and FTP server for centralized file sharing.
- Provision of Internet access for internal users through an edge router using Network Address Translation (NAT), allowing private internal IP addresses to communicate with external networks securely.
- Configuration of a dynamic routing protocol, specifically Open Shortest Path First (OSPF), to ensure efficient and scalable routing within the campus backbone.
- Enforcement of network security policies using Access Control Lists (ACLs) to restrict unauthorized access to sensitive networks such as the Administration and Server VLANs.

The designed solution must closely resemble a real-world campus network and should be implemented and tested using Cisco Packet Tracer. Proper verification through connectivity testing, service access validation, and security enforcement is required to demonstrate the successful fulfilment of all objectives.

3. Methodology

I. Network Design

- **Architecture:** Three-tier hierarchical model (Core, Distribution, Access)
- **Core Layer:** Central routing and backbone connectivity
- **Distribution Layer:** Redundancy and aggregation
- **Access Layer:** End-user connectivity



II. VLAN Creation

```
Switch>enable
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname SW-CORE
SW-CORE(config)#vlan 10
SW-CORE(config-vlan)# name CSE
SW-CORE(config-vlan)#exit
SW-CORE(config)#
SW-CORE(config)#vlan 20
SW-CORE(config-vlan)# name ECE
SW-CORE(config-vlan)#exit
SW-CORE(config)#
SW-CORE(config)#vlan 30
SW-CORE(config-vlan)# name MECH
SW-CORE(config-vlan)#exit
SW-CORE(config)#
SW-CORE(config)#vlan 40
SW-CORE(config-vlan)# name CIVIL
SW-CORE(config-vlan)#exit
SW-CORE(config)#
SW-CORE(config)#vlan 50
SW-CORE(config-vlan)# name ADMIN
SW-CORE(config-vlan)#exit
SW-CORE(config)#
SW-CORE(config)#vlan 60
SW-CORE(config-vlan)# name LIB
SW-CORE(config-vlan)#exit
SW-CORE(config)#
SW-CORE(config)#vlan 70
SW-CORE(config-vlan)# name SERVER
SW-CORE(config-vlan)#exit
SW-CORE(config)#
SW-CORE(config)#vlan 99
SW-CORE(config-vlan)# name MGMT
SW-CORE(config-vlan)#exit
SW-CORE(config)#
SW-CORE(config)#end
SW-CORE#write memory
%SYS-5-CONFIG_I: Configured from console by console
```

VLAN ID	VLAN Name
10	CSE
20	ECE
30	MECH
40	CIVIL
50	ADMIN
60	LIB
70	SERVER

VLAN TABLE

VLANs are used to logically segment the university network based on departments and services. This segmentation improves security, reduces broadcast traffic, and enhances overall network performance. Each VLAN is assigned a unique IP subnet, and inter-VLAN communication is enabled through the core routing device with appropriate security controls.

VLAN configuration:

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 50
Switch(config-if)#exit
Switch(config)#end
Switch#wr
Building configuration...
[OK]
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Switch#
%SYS-5-CONFIG_I: Configured from console by console
conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range fa0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 40
Switch(config-if-range)#exit
Switch(config)#end
Switch#wr
%SYS-5-CONFIG_I: Configured from console by console
```

```
SW-CORE>enable
SW-CORE#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-CORE(config)#interface range fa0/5 - 9
SW-CORE(config-if-range)# switchport mode access
SW-CORE(config-if-range)# switchport access vlan 70
SW-CORE(config-if-range)#exit
SW-CORE(config)#end
SW-CORE#wr
Building configuration...
[OK]
SW-CORE#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range fa0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#end
Switch#wr
Building configuration...
[OK]
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range fa0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#end
Switch#wr
Building configuration...
[OK]
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface fa0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 60
Switch(config-if)#exit
Switch(config)#end
Switch#wr
Building configuration...
[OK]
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

```
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range fa0/1 - 2
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)#exit
Switch(config)#end
Switch#wr
%SYS-5-CONFIG_I: Configured from console by console
```

III. IP Addressing Scheme

Each VLAN is assigned a /24 subnet:

- CSE: 192.168.10.0/24
- ECE: 192.168.20.0/24
- MECH: 192.168.30.0/24
- CIVIL: 192.168.40.0/24
- ADMIN: 192.168.50.0/24
- LIB: 192.168.60.0/24

DHCP Pool Configuration:

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server
LIB	192.168.60.1	192.168.70.3	192.168.60.10	255.255.255.0	50	0.0.0.0
ADMIN	192.168.50.1	192.168.70.3	192.168.50.10	255.255.255.0	50	0.0.0.0
CIVIL	192.168.40.1	192.168.70.3	192.168.40.10	255.255.255.0	100	0.0.0.0
MECH	192.168.30.1	192.168.70.3	192.168.30.10	255.255.255.0	100	0.0.0.0
ECE	192.168.20.1	192.168.70.3	192.168.20.10	255.255.255.0	100	0.0.0.0
CSE	192.168.10.1	192.168.70.3	192.168.10.10	255.255.255.0	100	0.0.0.0

IV. Routing and NAT

Routing plays a crucial role in enabling communication between different VLANs and network segments within the university campus. In this project, the Open Shortest Path First (OSPF) dynamic routing protocol is configured between the Core router and the Edge router to ensure efficient route exchange and fast convergence. OSPF allows the network to scale easily and automatically adapt to topology changes without manual reconfiguration.

To provide Internet connectivity to internal users, Network Address Translation (NAT) with Port Address Translation (PAT) is configured on the Edge Router. NAT enables devices using private IP addresses within the campus network to access external networks securely by translating them into a public IP address. This approach conserves public IP addresses and

adds an additional layer of security by hiding internal network details.

- **OSPF configured between Core and Edge routers**
- **NAT/PAT configured on Edge Router for Internet access**

R1-EDGE config

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# no shutdown

Router(config-if)#exit
Router(config)#end
Router#wr
Building configuration...
[OK]
Router#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

%SYS-5-CONFIG_I: Configured from console by console
```

R1-EDGE NAT config

```
Router>enable
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#interface g0/1
Router(config-if)# description TO_R2_CORE
Router(config-if)# ip address 10.0.0.1 255.255.255.252
Router(config-if)# ip nat inside
Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#interface g0/0
Router(config-if)# description TO_INTERNET
Router(config-if)# ip nat outside
Router(config-if)# no shutdown

Router(config-if)#exit
Router(config)#
Router(config)#end
Router#wr
Building configuration...
[OK]
Router#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

R2-CORE config

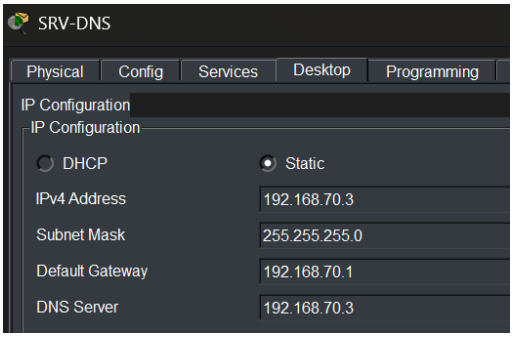
```
R2-CORE#enable
R2-CORE#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2-CORE(config)#interface g0/1
R2-CORE(config-if)# ip address 192.168.99.2 255.255.255.0
R2-CORE(config-if)# no shutdown

R2-CORE(config-if)#exit
R2-CORE(config)#end
R2-CORE#wr
Building configuration...
[OK]
R2-CORE#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

V. Server Configuration

The server farm plays a central role in the university campus network by hosting all essential network services required for day-to-day operations. All servers are placed within a dedicated Server VLAN to ensure better security, centralized management, and controlled access from other departments. Assigning static IP addresses to servers ensures reliability, ease of troubleshooting, and consistent service availability across the network.

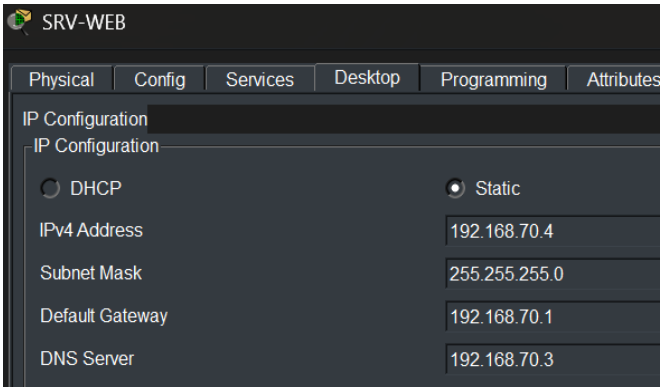
Service	IP address
DHCP	192.168.70.2
DNS	192.168.70.3
WEB	192.168.70.4
MAIL	192.168.70.5
FTP	192.168.70.6

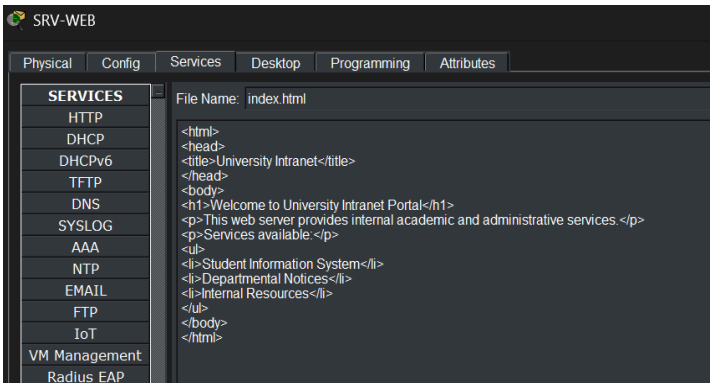
DNS Records Configuration



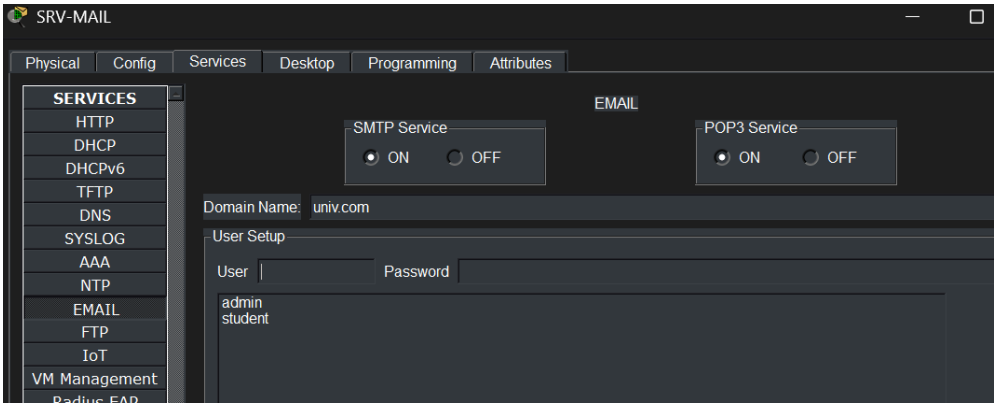
No.	Name	Type	Detail
0	dhcp.univ.com	A Record	192.168.70.2
1	dns.univ.com	A Record	192.168.70.3
2	ftp.univ.com	A Record	192.168.70.6
3	mail.univ.com	A Record	192.168.70.5
4	web.univ.com	A Record	192.168.70.4

Web Server Configuration and Intranet Page

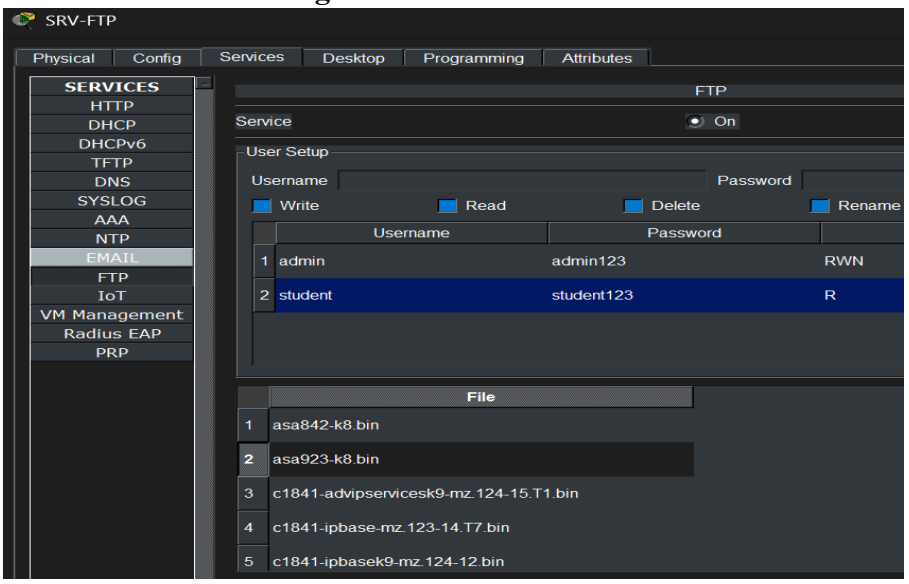




Email Server Configuration



FTP Server User Configuration



VI. Security (ACLs)

Security is a critical aspect of any campus network, especially in an environment where multiple departments and users share common infrastructure. In this project, Access Control Lists (ACLs) are implemented to enforce security policies and restrict unauthorized access between different VLANs. ACLs help in controlling traffic flow based on source IP address, destination IP address, and protocol type, thereby protecting sensitive network resources.

The **Administration VLAN** is isolated from student VLANs to prevent unauthorized access to confidential administrative data. Similarly, the **Server VLAN** is secured using extended ACLs so that only permitted departments can access specific services such as web, mail, or FTP. This ensures that critical servers remain protected from misuse or potential attacks originating from user networks.

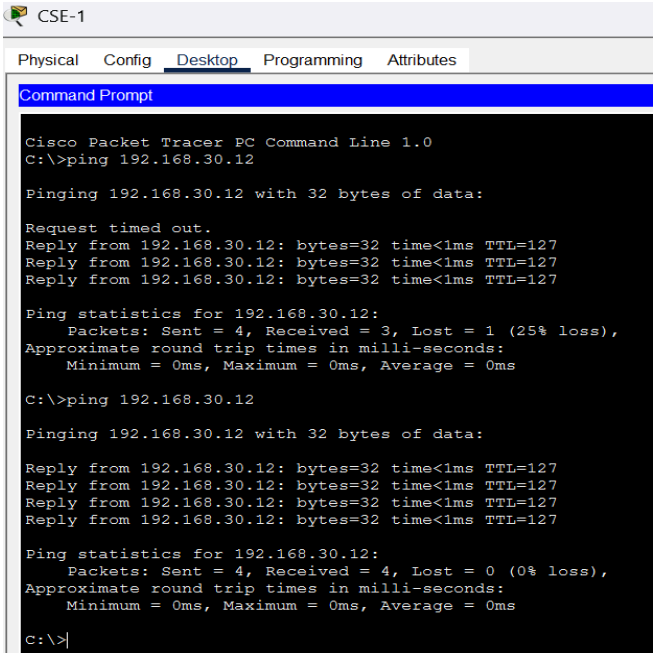
- ADMIN VLAN restricted from student VLANs
- Server VLAN protected using extended ACLs

```
SW-CORE#enable
SW-CORE#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-CORE(config)#
SW-CORE(config)#ip access-list extended BLOCK_ADMIN
SW-CORE(config-ext-nacl)# deny ip 192.168.10.0 0.0.0.255 192.168.50.0 0.0.0.255
SW-CORE(config-ext-nacl)# deny ip 192.168.20.0 0.0.0.255 192.168.50.0 0.0.0.255
SW-CORE(config-ext-nacl)# deny ip 192.168.30.0 0.0.0.255 192.168.50.0 0.0.0.255
SW-CORE(config-ext-nacl)# deny ip 192.168.40.0 0.0.0.255 192.168.50.0 0.0.0.255
SW-CORE(config-ext-nacl)# deny ip 192.168.60.0 0.0.0.255 192.168.50.0 0.0.0.255
SW-CORE(config-ext-nacl)# permit ip any any
SW-CORE(config-ext-nacl)#exit
SW-CORE(config)#exit
SW-CORE#
%SYS-5-CONFIG_I: Configured from console by console

SW-CORE#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-CORE(config)#ip access-list extended BLOCK_SERVERS
SW-CORE(config-ext-nacl)# deny ip 192.168.10.0 0.0.0.255 192.168.70.0 0.0.0.255
SW-CORE(config-ext-nacl)# deny ip 192.168.20.0 0.0.0.255 192.168.70.0 0.0.0.255
SW-CORE(config-ext-nacl)# deny ip 192.168.30.0 0.0.0.255 192.168.70.0 0.0.0.255
SW-CORE(config-ext-nacl)# deny ip 192.168.40.0 0.0.0.255 192.168.70.0 0.0.0.255
SW-CORE(config-ext-nacl)# deny ip 192.168.60.0 0.0.0.255 192.168.70.0 0.0.0.255
SW-CORE(config-ext-nacl)# permit ip any any
SW-CORE(config-ext-nacl)#exit
SW-CORE(config)#interface vlan 50
SW-CORE(config-if)# ip access-group BLOCK_ADMIN in
SW-CORE(config-if)#exit
SW-CORE(config)#end
SW-CORE#wr
Building configuration...
```

4. Results & Interpretation

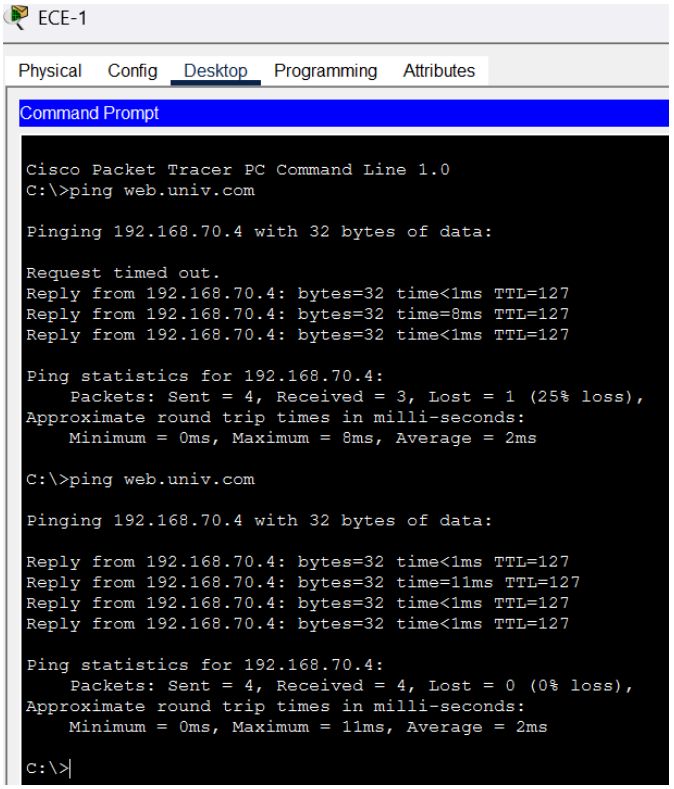
I. Connectivity Testing

- Successful ping between VLANs from CSE department to MECH department.

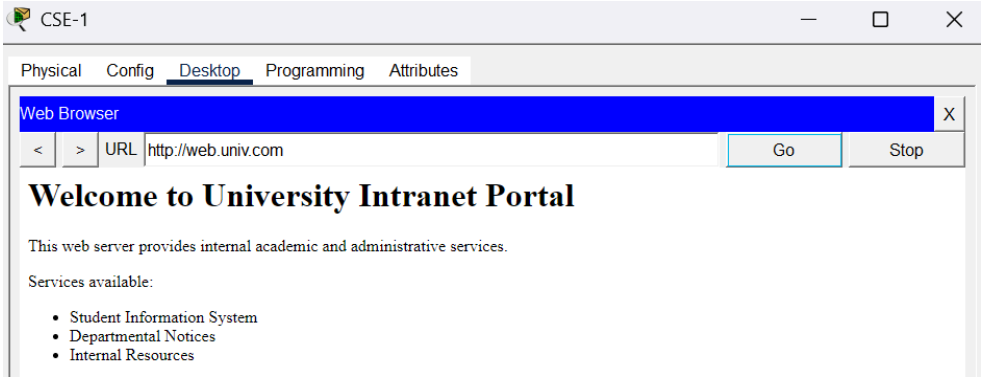


Ping between VLAN 10 to VLAN 30

- DNS name resolution verified



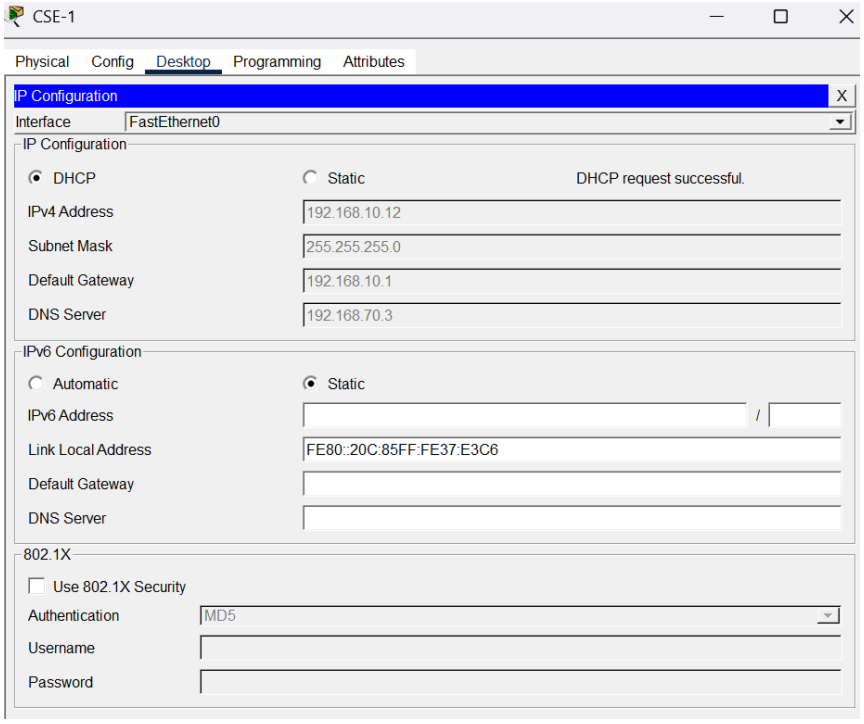
DNS Resolution (ping/web using domain name)



Web access via http://web.univ.com confirmed

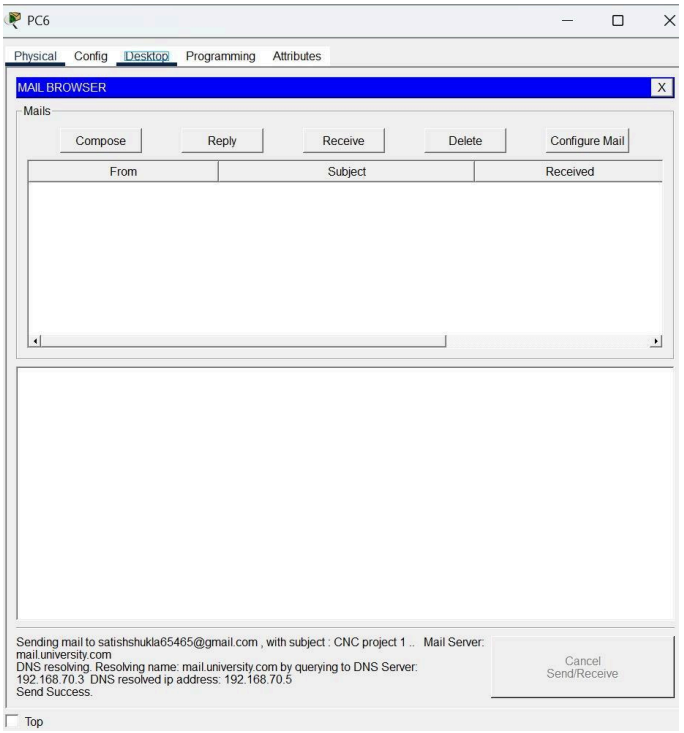
II. Service Verification

- DHCP dynamically assigns IP addresses



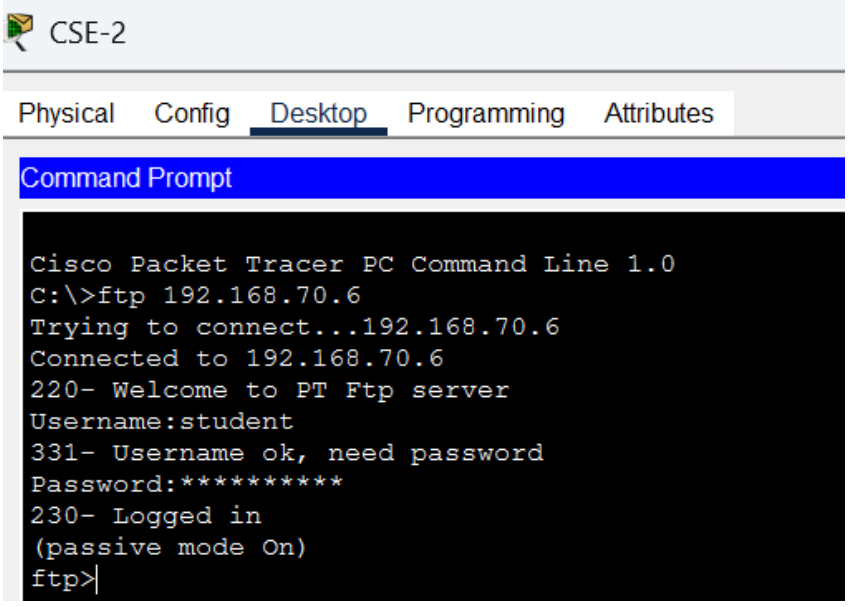
DHCP request successful message

- Email service verified using SMTP and POP3



Mail Exchange

- FTP service authenticated using user accounts



FTP Service Authenticated Using User Accounts

5. Conclusion

The university campus network was successfully designed and implemented using Cisco Packet Tracer by following a structured and hierarchical approach that closely aligns with real-world enterprise and campus networking practices. The project effectively demonstrates the practical application of core computer networking concepts such as VLAN-based segmentation, inter-VLAN routing, dynamic routing using the Open Shortest Path First (OSPF) protocol, centralized server deployment, Network Address Translation (NAT), and Access Control Lists (ACLs) for enforcing security policies.

By logically segmenting departments and services into separate VLANs, the network design significantly reduces unnecessary broadcast traffic and enhances overall network performance. This structured segmentation also improves security by isolating sensitive departments such as Administration and critical resources such as the Server Farm from general user access. The implementation of inter-VLAN routing ensures seamless and controlled communication between departments while maintaining logical separation.

The successful configuration of essential network services including DHCP, DNS, Web, Email, and FTP servers highlights the importance of centralized service management in a campus environment. Dynamic IP address allocation through DHCP simplifies network administration, while DNS improves usability by enabling domain-based access to services. The availability of Web, Email, and FTP services demonstrates effective resource sharing and internal communication across the university network.

Internet connectivity for internal users is achieved through the implementation of NAT on the edge router, allowing private IP addresses to securely access external networks while conserving public IP resources. Additionally, the use of ACLs provides an added layer of protection by restricting unauthorized access and ensuring that only permitted traffic is allowed to reach sensitive VLANs and services.

Overall, this project closely resembles a real-world university or enterprise network and provides valuable hands-on experience in network planning, configuration, testing, and troubleshooting. It strengthens the understanding of campus network architecture, improves practical knowledge of networking devices and protocols, and prepares students for real-world networking scenarios, professional certifications, and advanced studies in computer networking

6. References

[1] CompTIA Network+ N10-008 Certification Guide by Glen D. Singh, 2*nd* Edition, Packt publication.

[2] Cisco Networking Academy Documentation.
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